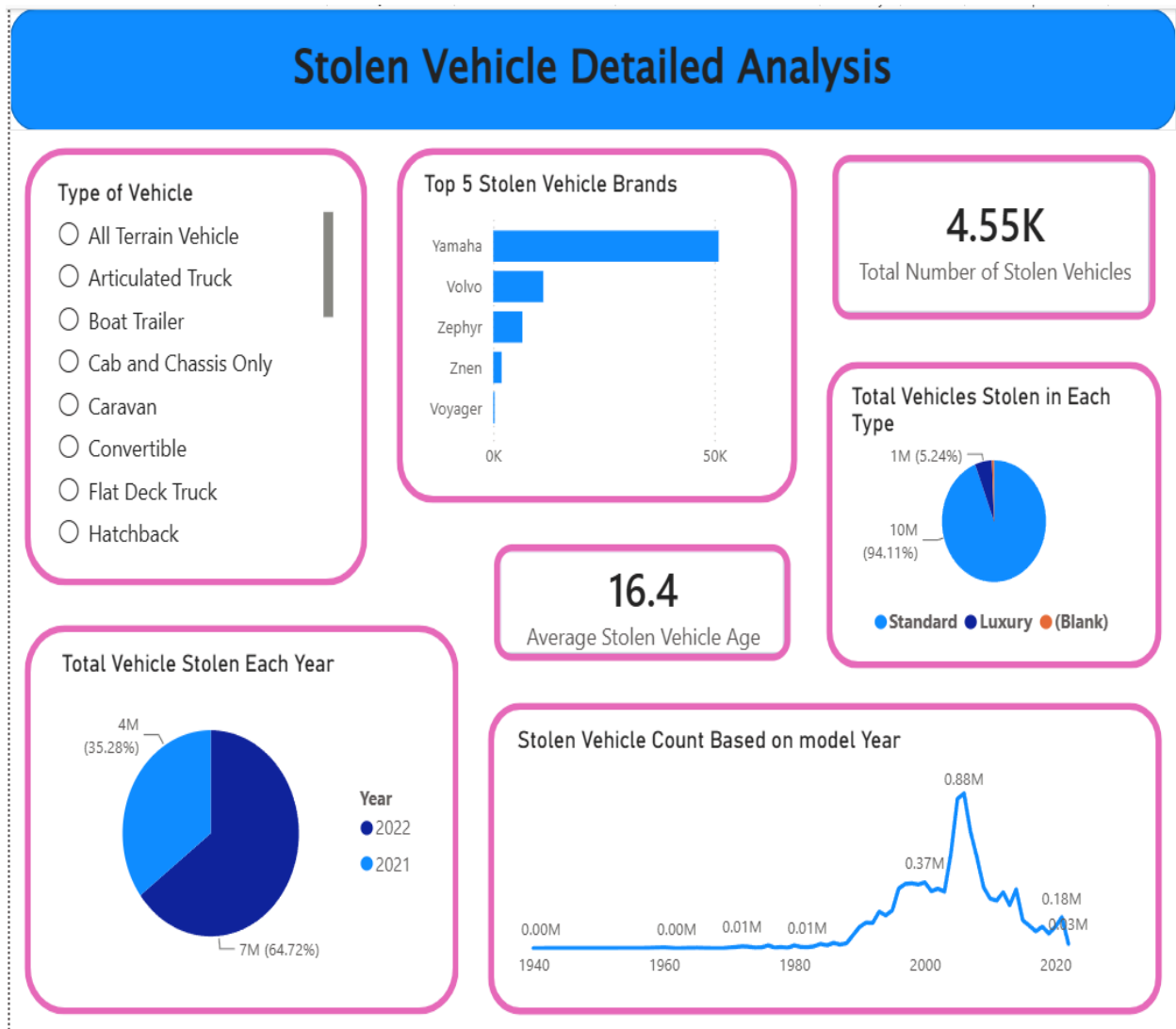


VEHICLE THEFT ANALYSIS REPORT (2021–2022)

Prepared for: Stakeholder Presentation

Objective: To identify vehicle theft patterns, evaluate Year-over-Year (YoY) growth, profile high-risk assets, and provide actionable security recommendations.



1. Executive Summary

An analysis of vehicle theft records across a two-year period reveals a sharp, critical escalation in theft incidents. The data identifies a highly concentrated risk profile: thieves disproportionately target standard-class utility vehicles and older models manufactured in the late 1990s and early 2000s. This pattern points strongly toward vulnerabilities in legacy security systems as the primary driver of successful thefts.

2. Key Performance Indicators (KPIs)

The high-level metrics establish a baseline understanding of the volume and physical traits of targeted vehicles:

- **Total Volume:** 4.55K total vehicles recorded as stolen over the tracked period.
- **Average Vehicle Age:** 16.4 years at the time of theft, demonstrating a clear preference for older assets.
- **Segment Domination:** **Standard Class** vehicles account for **94.11%** of all thefts, whereas **Luxury Class** assets make up only **5.24%**.

3. Temporal Trends & Year-over-Year (YoY) Growth

Theft Volume by Calendar Year

Theft activity accelerated dramatically between the two observed tracking periods:

Tracking Year	Stolen Volume	Share of Total Theft
2021	1.67K	35.3%
2022	2.89K	64.7%

Key Observation: Vehicle thefts surged by approximately **73%** in 2022 compared to 2021. This sharp upward trajectory indicates an escalating systemic issue rather than a stable, predictable baseline.

Vulnerability by Vehicle Model Year

The historical manufacturing year of the vehicle is one of the strongest predictors of theft risk:

- **The 2000 Peak:** The volume of thefts forms a classic bell curve that peaks sharply around vehicles manufactured between **1998 and 2005**.
- **The Legacy Security Gap:** Vehicles built around the year 2000 experience the highest absolute volume of incidents. Conversely, modern vehicles (Model Years 2015–2022) show minimal theft volumes, confirming that modern anti-theft tech (transponder keys, engine immobilizers) acts as an effective deterrent.

4. Vehicle Profile Risk Analysis

Top 5 Brands Most Targeted (Overall Fleet)

When looking at the entire pool of stolen vehicles, the risk is heavily concentrated among specific manufacturers:

- 1. **Yamaha** (Highest volume by a significant margin, suggesting extreme vulnerability in the motorcycle/powersports category)
- 2. **Volvo**
- 3. **Zephyr**
- 4. **Znen**
- 5. **Voyager**

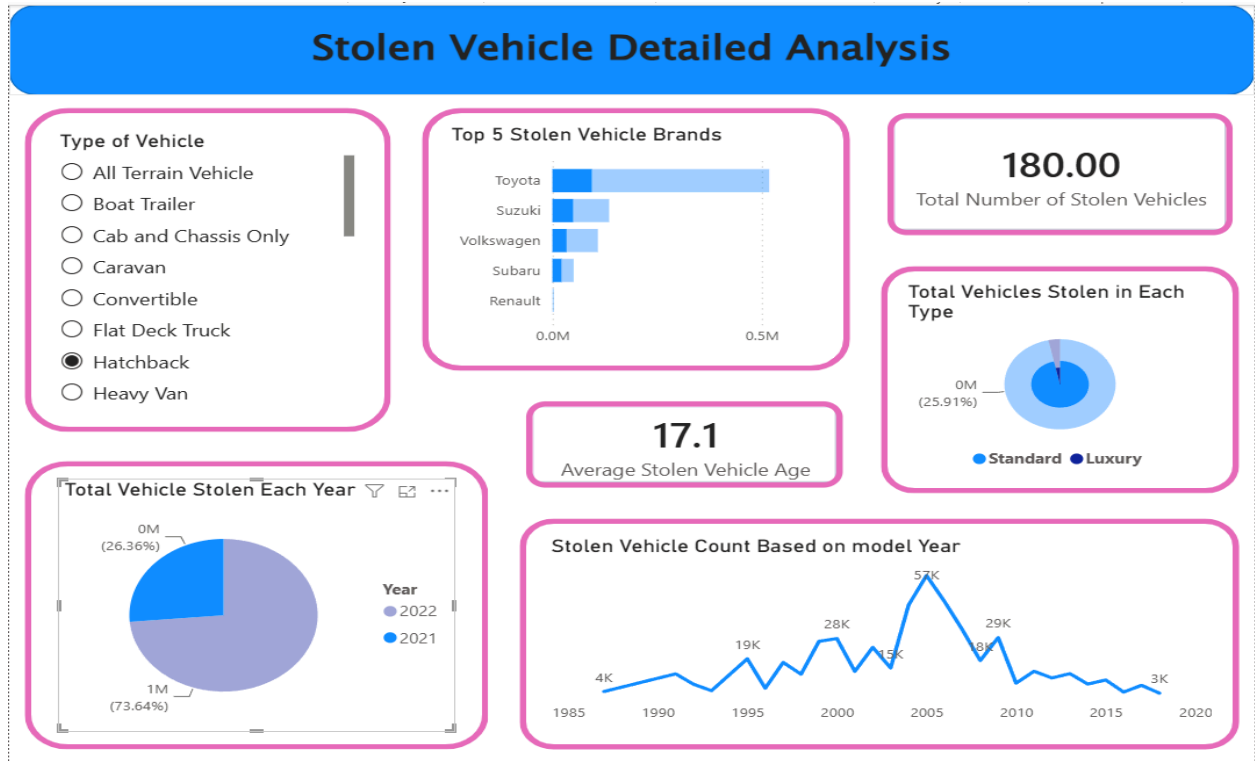
5. Segment Spotlight: Hatchbacks vs. Overall Fleet

Filtering the data down to specific body types reveals how consumer passenger vehicles differ from the broader powersport/commercial baseline.

Metric	Overall Fleet Baseline	Hatchback Segment
Total Volume	4,550 units	180 units
Average Age	16.4 years	17.1 years
Peak Model Year Range	1998–2001	2002–2005
Primary Target Brands	Yamaha, Volvo, Zephyr	Toyota, Suzuki, Volkswagen, Subaru, Renault

Key Segment Insights:

- **Brand Shift:** When removing powersports/motorcycles, **Toyota** and **Suzuki** emerge as the most vulnerable brands for standard passenger hatchbacks.
- **Delayed Peak:** Stolen hatchbacks are, on average, slightly older (17.1 years) than the general fleet, with the risk window shifting slightly right to the 2002–2005 manufacturing bracket.



6. Strategic Recommendations

For Law Enforcement & Municipalities

- **Targeted Anti-Theft Campaigns:** Launch public awareness campaigns focusing on owners of early-2000s models (particularly Toyota, Suzuki, and Yamaha owners), educating them on physical security options.
- **Geographic & Market Interdiction:** The high volume of Yamaha thefts suggests an active market for parts or specialized transport. Focus investigative efforts on specialized chop shops and secondary markets dealing in powersport components.

For Insurance Providers & Risk Management

- **Premium Incentives for Retrofitting:** Offer premium discounts or incentives for policyholders who install aftermarket security layers (such as GPS trackers, steering wheel clubs, or fuel pump kill switches) on vehicles manufactured prior to 2006.
- **Adjusted Risk Models:** Update actuarial models to reflect the 73% year-over-year escalation in theft frequency, ensuring risk profiles are adjusted for legacy standard-class assets.